Jacob Miller

Curriculum Vitae

Postdoctoral Fellow – Technical University of Munich
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Google Scholar DBLP

EDUCATION

2024 **PhD**, Computer Science. Minor: Mathematics. University of Arizona.
2020 **BS**, Computer Science. *Summa Cum Laude.* Southwestern Oklahoma State University.

EXPERIENCE

2024– Postdoctoral Fellow, Technical University of Munich
2022–2024 PhD Intern, Pacific Northwest National Laboratory
2021–2024 Research/Teaching Assistant, University of Arizona
2020 Modeling and Simulation Intern, Federal Aviation Administration
2019 Modeling and Simulation Intern, National Aeronautics and Space Administration

PUBLICATIONS

Journal Publications

1. **Miller, Jacob**, Markus Wallinger, Ludwig Felder, Timo Brand, Henry Förster, Johannes Zink, Chunyang Chen, and Stephen Kobourov. "Exploring MLLMs Perception of Network Visualization Principles." *IEEE Transactions on Visualization and Computer Graphics. Preprint.* 2026.
2. Smelser, Kiran, Kaviru Gunaratne, **Jacob Miller**, and Stephen Kobourov. "How Scale Breaks "Normalized Stress" and KL Divergence: Rethinking Quality Metrics." *IEEE Transactions on Visualization and Computer Graphics. Preprint.* 2026.
3. Efrat, Alon, William Evans, Kassian Köck, Stephen Kobourov, **Jacob Miller**. "Visualization of Bipartite Graphs in Limited Window Size." *Acta Informatica 62,* 19. 2025.
4. **Miller, Jacob**, Dhruv Bhatia, Stephen Kobourov. "State of the Art of Graph Visualization in non-Euclidean Spaces." *EuroVis STAR, Computer Graphics Forum, 43*, 3, e15113. 2024.

Conference Publications

1. Ji, Ya, Xuefeng Li, Timo Brand, **Jacob Miller**, Peng Zhang, Stephen Kobourov, Yifan Hu. "Visualizing High-Dimensional Graph Embeddings via Informed Multi-View Projections." *IEEE VIS 2026* ***Best Paper Honorable Mention***
2. Gunaratne, Kaviru, Stephen Kobourov, **Jacob Miller**. "Class Angular Distortion Index for Dimensionality Reduction." *EuroVis 2026.*
3. Wallinger, Markus, **Jacob Miller**, Andrei Maftei, Stephen Kobourov. "Quantitative Metrics for Edge Bundling of Network Visualizations." *EuroVis 2026.* ***Best Paper Honorable Mention***
4. Felder, Ludwig, **Jacob Miller**, Markus Wallinger, Stephen Kobourov, Chunyang Chen. "ReTrace: Interactive Visualizations for Reasoning Traces of Large Reasoning Models." *EuroVis 2026.*
5. Mooney, Gavin J., **Jacob Miller**, Michael Wybrow, Stephen Kobourov, Helen Purchase. "Stress in Graph Drawings: Perception, Preference, and Performance." *International Symposium on Graph Drawing and Network Visualization.* 2025.
6. **Miller, Jacob**, Dhruv Bhatia, Helen Purchase, Stephen Kobourov. "Euclidean, Hyperbolic, and Spherical Networks: An Empirical Study of Matching Network Structure to Best Visualizations." *EuroVis 2025.*
7. Mooney, Gavin J., Helen Purchase, Michael Wybrow, Stephen Kobourov, and **Jacob Miller**. "The Perception of Stress in Graph Drawings." *International Symposium on Graph Drawing and Network Visualization.* 2024.
8. Mackey, Patrick, **Jacob Miller**, Liz Faultersack. "Improving Property Graph Layouts by Leveraging Attribute Similarity for Structurally Equivalent Nodes." *2024 IEEE VIS Short Papers.*
9. **Miller, Jacob**, Vahan Huroyan, Raymundo Navarrete, Md Iqbal Hossain, Stephen Kobourov. "ENS-t-SNE: Embedding Neighborhoods Simultaneously t-SNE." *2024 IEEE 17th Pacific Visualization Symposium (PacificVis).* IEEE, 2024.
10. **Miller, Jacob**, Vahan Huroyan, and Stephen Kobourov. "Balancing Between the Local and Global Structures (LGS) in Graph Embedding." *International Symposium on Graph Drawing and Network Visualization.* Cham: Springer Nature Switzerland, 2023.
11. **Miller, Jacob**, Mohammad Ghoniem, Hsiang-Yun Wu, Helen Purchase. "On the Perception of Small Sub-graphs." *International Symposium on Graph Drawing and Network Visualization.* Cham: Springer Nature Switzerland, 2023.

12. **Miller, Jacob**, Vahan Huroyan, and Stephen Kobourov. "Spherical Graph Drawing by Multi-dimensional Scaling." *International Symposium on Graph Drawing and Network Visualization*. Cham: Springer International Publishing, 2022.
13. **Miller, Jacob**, Stephen Kobourov, and Vahan Huroyan. "Browser-based hyperbolic visualization of graphs." *2022 IEEE 15th Pacific Visualization Symposium (PacificVis)*. IEEE, 2022.

Workshop Publications

1. Hangan, Daniel, Stephen Kobourov, **Jacob Miller**. "Bridging Dimensionality Reduction and Graph Drawing with Stochastic Stress Optimization." *GDxDR Workshop 2026 (co-located with EuroVis)*.
2. Brand, Timo, Henry Förster, Stephen Kobourov, **Jacob Miller**. "Visualization Biases MLLM's Decision Making in Network Data Tasks." *VisXGenAI Workshop 2025 (co-located with IEEE VIS)*.
3. Smelser, Kiran, **Jacob Miller**, Stephen Kobourov. "'Normalized Stress' is Not Normalized: How to Interpret Stress Correctly." *2024 BELIV Workshop*.

TOOLS

- Dagstuhl Network Tool
- GMAP: Under advanced options → spherical/hyperbolic
- Spherical Multidimensional Scaling (SMDS)
- Spherical and hyperbolic graph visualizations
- ENS-t-SNE

Funding

Funded

Proposal Title: Bridging Graph Drawing and Dimensionality Reduction
Funding Agency: TUM Incentive Fund
Role Status: PI
Total direct costs: €97,000
Coverage Period: Oct 2026 – Sep 2027.

Pending

Proposal Title: GALA - Graph Analysis with LLM Assistance
Funding Agency: DFG
Role Status: Project Group Leader
Total direct costs: €1,132,529

SERVICE

- PC member – IEEE PacificVis (2027)
- PC member – SOFSEM (2027)
- PC member – IEEE VIS Short Papers (2026)
- PC member – GDxDR Workshop (associated with EuroVis) (2026)
- PC member – EuroVA Workshop (associated with EuroVis) (2026)
- PC member – EuroVis Posters and Demos (2026)
- Co-chair – ETVIS Workshop (associated with ETRA) (2026)
- PC member – EuroVis Posters and Demos (2025)

TEACHING

- Bachelor/Master Seminar on Algorithm Engineering – Coordinated and advised student research projects
- Bachelor/Master Practical on Algorithm Competitive Challenges – Advised student projects
- Data Science (Bachelor) – Created and taught lecture content and exercises
- Data Structures and Algorithms (Bachelor) – Coordinated 5 weekly tutorial sessions and taught 2 of them
- Introduction to Data Visualization (Bachelor) – Created and taught lecture content and exercises
  - Student project received an honorable mention in the International VAST Challenge competition.
- Theory of Computation (Bachelor) – Coordinated 5 weekly tutorial sessions and taught 3 of them